

A memory-based learning approach utilizing combined spectral sources and geographical proximity for improved VIS-NIR-SWIR soil properties estimation

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ABSTRACT

The current study was driven by the need to derive improved soil information whereby the strengths of memory-based learning and soil spectroscopy are exploited towards addressing regional challenges and supporting sustainable development across the Balkan, North Africa, and Middle East regions. In this study we focused on a novel derivation of the Spectrum Based Learner (SBL) algorithm by i) taking into account both the geographical proximity and the spectral similarity in the computation of the distance between samples, and ii) using an optimal pair of spectral pre-treatments with the first used for the determination of neighbours while the latter for the estimation of the target property. The proposed approach was tested on a recently developed standardized Soil Spectral Library (SSL) across the VIS-NIR-SWIR spectral region range (350–2500 nm) which comprises 1760 soil samples from 9 different countries and is considered the largest and most diverse soil database infrastructure in the region compatible with other SSLs. Here, we found that our approach presents a great potential for predicting soil texture contents (Clay, Sand, and Silt), organic carbon (OC), Calcium Carbonate (CaCO₃), pH, and Electrical Conductivity (EC). These results outperform the predictive performance of other state of the art global and local approaches that have been applied to similar large and complex soil datasets. The use of geographical coordinates in the computation of the sample similarities enhanced the predictions of soil properties, since it allowed the generation of local subsets that present similar soil compositions. In addition, we conclude that the various spectral sources as derived by a set of predefined spectral pre-processing techniques contain complementary information which should be combined instead of relying solely on the best spectral pre-processing technique. This approach could be effectively utilized to enhance the predictions of soil properties in large and complex SSLs, since it decreased the Root Mean Square Errors (RMSEs) by a relative mean of 6.47% (average value across the properties – decrease ranging from 2.90% to 9.09%) for the various soil properties, compared to other global and local algorithms. We conclude that national SSLs that were measured under a standardization process could further contribute to the global initiative to address challenges and support a data-centric approach for informed decision making with regards to environmental and agricultural issues.

1. Introduction

Soil functions in the Balkan, North Africa, and Middle East region, as in many parts of Mediterranean, are under increasing pressure of intensification and competing land uses. This leads to soil erosion, nutrient imbalance and loss of OC which threaten the sustainable management of soil resources (Montanarella et al., 2016). In that

context, there is a growing realisation among policy-makers and scientists that a systematic approach for soil monitoring is key to resolving the aforementioned challenges, support a number of policies, and identify opportunities to reach the “4 per mille Soils for Food Security and Climate” components as introduced in the 21st Conference of Parties to the United Nations Framework Convention on Climate Change.

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Over the past few decades, it is widely stated that VIS-NIR-SWIR spectroscopy holds a prominent role to enhance our ability to measure and monitor soil's ecosystem components, since it is recognized as a novel, efficient and cost effective approach compared to the conventional laboratory methods (Nocita et al., 2015). In this context, several novel and potentially low-cost in situ observational platforms (Viscarra Rossel et al., 2011) are rapidly maturing and also considered viable alternatives to analytical chemistry methods (O'Rourke and Holden, 2011). A wide range of activities have witnessed VIS-NIR-SWIR spectroscopy as a key enabler for the qualitative and quantitative description of soils in the laboratory and in real field conditions. These range from small experiments in individual fields (Bartholomeus et al., 2011), to regions and countries (Nocita et al., 2011), to large scale continental, such as the European LUCAS top soil database (Tóth et al., 2013) as well as its expansion to further cover geographically areas with well reported gaps such as the Balkans (Orgiazzi et al., 2017), and global surveys (Viscarra Rossel et al., 2016).

The above cases have utilized several robust methods to characterize the chemical, physical and mineralogical composition of soil. However, developing accurate spectroscopic models that use large SSLs still remains a challenging task. This characteristic is amplified in cases where libraries comprise diverse samples in terms of geographical origin, parent material, measurement protocols and environmental conditions (Guerrero et al., 2010). Earlier studies indicated that advanced local regression approaches enable progress beyond the state-of-the-art in terms of prediction accuracy. For instance, Genot et al. (2011) utilized the LOCAL algorithm (Shenk et al., 1997) for predicting soil attributes using a regional NIR soil library, while Ramirez-Lopez et al. (2013) showed using a memory based learner for modelling soil spectra across the VIS-NIR-SWIR spectral region that their local algorithm combined with soil distance matrices performed better than commonly employed machine learning models. In similar manner, Nocita et al. (2014) demonstrated that incorporating additional information, such as geographical and texture elements, in the computation of the distance between samples in a local regression procedure, further improved the prediction performance of the spectral models. Similarly, Shi et al. (2015) showed that the use of a spatially constrained local Partial Least Squares (PLS) regression based on soil geographical information substantially increased the predictive power of models in complex datasets.

While these activities have made great strides towards developing novel regression techniques, and soil-spectral data archives, the development of SSLs is still not adequately taken up or is completely lacking, in some Balkan countries (Orgiazzi et al., 2017) and in Middle East, but even more so in North Africa (Viscarra Rossel et al., 2016). In recognition of the inhomogeneous picture of the region's contribution in the various continental and global mapping activities, spanning from the absolute absence to representation of many countries with only few soil spectra, we focused on providing a concrete contribution to cover these recognized gaps within the framework of the GEO-CRADLE project (Coordinating and integrating state-of-the-art Earth Observation activities in the regions of North Africa, Middle East, and Balkans and developing links with GEO related initiatives towards Global Earth Observation System of Systems-GEOSS). Hence, we developed a diverse database of quality, open and disaggregated soil VIS-NIR-SWIR spectral data in the region. Compared to similar international efforts, as the global SSL, we focussed, for the first time, on implementing a standard measurement protocol in order to yield better data merging capability and analyses and to minimize the spectral measurement variations in these diverse and big spectroscopic data. Previous studies have highlighted the need for the application of standard measurement protocols and internal soil standard methods (Ben Dor et al., 2015) between collaborating actors, to optimally allow the processing of the collected set of heterogeneous data due to diverse sources and spectrometers, towards the enhancement of predictive performance (Romero et al., 2018). Taking into account the above, we promote the development of a SSL as a part of the long-term implementation of a Global SSL

enabling the further integration of untapped national soil archives into one harmonized soil database. In this line, it should be mentioned that other initiatives, such as the LUCAS SSL, achieved to develop a significantly large, comprehensive and harmonized soil database by analysing all samples in a single laboratory. To that end, we developed the GEO-CRADLE SSL, composed of 1760 samples from 1017 soil profiles distributed in nine countries from three continents and covering an area with well recognized gaps in soil data. The samples were selected from national soil data archives or collected through field surveyors during the project's lifetime, following general guidelines, standards, and protocols to ensure consistent data collection and analysis. This is in agreement with recent studies in which the effective integration of Earth observation capacities and the uptake of soil derived outputs in response to informed decision making with regards to environmental and agricultural issues are highlighted (Castaldi et al., 2018).

In this study, our goal was to report the development of a new memory-based learning approach named LGR (Local Gaussian Regression). It was designed to work by combining the geographical proximity and the spectral similarity in order to derive distance matrices for the determination of neighbours and use as additional predictors. This work extends previous works on this domain by proposing for the first time the use of combined spectral sources, whereby two different spectral pre-treatments are used, one for the spectral similarity and the other to develop predictors for the target soil property. Our hypothesis is that the complementary information contained among different spectral sources as derived by different pre-processing techniques will enhance the prediction capabilities of the models. Hence, one spectral source can be used to improve the knowledge at the selection of representative local subsets and a different one can be utilized to improve the predictive performance of the local models. The performance of LGR was evaluated in the recently developed GEO-CRADLE SSL across the Balkan, Middle East and North Africa region, presented here for the first time.

2. Materials and methods

2.1. The GEO-CRADLE survey

The GEO-CRADLE project has brought together nine national research organizations from three continents, to develop a regional data infrastructure of soil VIS-NIR-SWIR spectra across the range of 350 to 2500 nm. In this line, the GEO-CRADLE SSL includes 1760 geo-referenced soil samples from nine countries across the Balkan, Middle East and North Africa region (Fig. 1). Approximately 58% of the spectra belong to the top soil layer (0–30 cm), 20% and 20% of the samples collected at depths within 30–60 cm and 60–100 cm, respectively, while the 2% originate from samples collected at depths > 100 cm.

In the spring and summer of 2017, 393 points were selected and visited in the field for data collection where soil surveys were conducted in four countries, namely Albania (AL), the Former Yugoslav Republic of Macedonia (FY), Turkey (TR), and the Republic of Serbia (RS). Bulgaria (BG) was surveyed in autumn 2017. The soil samples in the Greek SSL (GR) are historical samples which were collected for soil survey purposes in different croplands located in the region of the Eastern Macedonia and Thrace in 2014 (Tsakiridis et al., 2017a).

GEO-CRADLE SSL also contains spectra with corresponding soil attribute data, from previously published local databases in Cyprus (CY; Cohen et al., 2011), and Israel (IS; Ben-Dor and Banin, 1994), while the Egyptian (EG) soil samples were collected in 2011, constrained in cropland areas.

The GEO-CRADLE survey aimed to implement an appropriate soil survey design allowing the most diverse utilisation of the results without compromising their scientific merit. In this context a stratified sampling scheme based on land use information was adopted, as previously implemented in Greek national soil monitoring scheme (EUDASM 2015). Since one of the objectives of the GEO-CRADLE

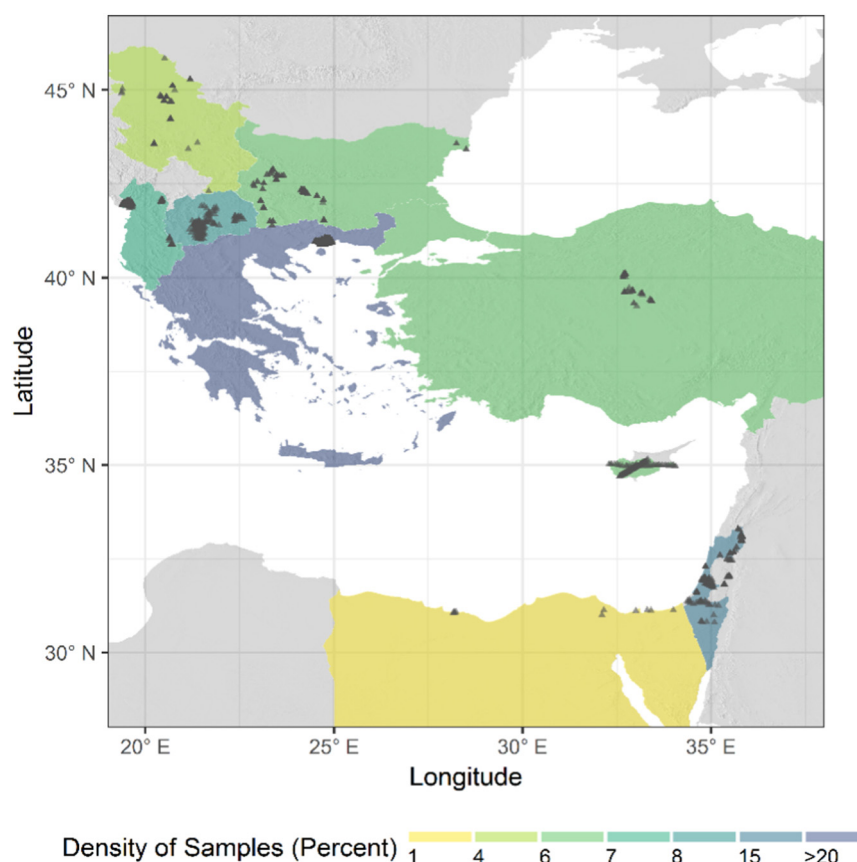


Fig. 1. Location of the 1760 sample sites with reflectance spectra in the GEO-CRADLE SSL.

survey was to collect information mainly related to croplands (areas where crops are planted and cultivated as referred to LUCAS sampling strategy), the Corine Land Cover 2012 dataset was used for constraining the area per region of interest for each country. These regions of interest were ranging from river basin level to entire countries. As additional information to the Corine Land Cover 2012, national land cover datasets were included. AL, FY, RS and BG have adapted the sampling methodology aforementioned.

The soil samples in GEO-CRADLE SSL were collected with geographical coordinates in the World Geodetic System (WGS84). The number of soil samples by country and the contributing organizations are listed in Table 1. The GEO-CRADLE SSL is compatible with GEOSS data principles and Open Database Licence standards and is available for download through the GEO-CRADLE datahub.¹

2.2. Soil sample analysis

The soil samples were air-dried, crushed, sieved and the fine earth (< 2.0 mm) was split into two equal parts. The first part was sent to the regional focal points' chemical laboratories which assessed the minimum number of soil parameters (OC, sand, silt and clay content) requested by the global SSL (Viscarra Rossel et al., 2016; see Appendix A), as well as parameters that were desirable with regards to national needs (CaCO₃, pH, EC). The second part of the soil sample was used for the spectral analysis.

2.2.1. Chemical laboratory analysis

The Walkley-Black method was applied for the measurement of OC (Heanes, 1984), while the concentrations of CaCO₃ were determined using a volumetric calcimeter. Soil clay-sand-silt contents were

Table 1

Number of spectra by country and the contributing organizations.

Country (abbreviation)	Contributor	N
Albania (AL)	Institute of Nature Conservation in Albania	107
Bulgaria (BG)	Bulgarian Academy of Sciences	105
Cyprus (CY)	Geological Survey Department	103
Egypt (EG)	Center for Environment and Development for the Arab Region and Europe	10
Greece (GR)	Interbalkan Environment Center	928
Israel (IL)	Tel Aviv University	221
Republic of Serbia (RS)	Institute of Physics Belgrade	63
FYROM (FY)	SS. Cyril and Methodius University in Skopje	124
Turkey (TR)	Scientific and Technological Research Council of Turkey	99

determined by the Bouyoucos (1962) method. The soil EC was determined with extracts of soil and water in 1:5 saturation (Bower and Wilcox, 1965) and pH measured in soil solution extracted from a mixture of 1:1 water/soil ratio (Peech, 1965).

The quality control of the soil analysis was secured by a quality assurance protocol promoted and adopted by all laboratories involved in several steps, including control of registration of samples, application of common calibration standards, repeated analyses of randomly selected samples and provision of randomly selected samples for comparison (10%) via an inter-calibration procedure with the certified laboratory of Interbalkan Environment Center. The Wilcoxon signed rank test was applied to ascertain whether there were any statistically significant differences between the measurements of the laboratories. Its results (results not shown) verified that both laboratories produced similar results, i.e. with no statistically significant difference.

¹ <http://datahub.geocradle.eu/dataset/regional-soil-spectral-library>.

2.2.2. VIS-NIR-SWIR measurement and processing

The national SSLs were created using two different spectrometers (350–2500 nm), an ASD Fieldspec Pro (PANalytical B-V, Boulder, CO, USA formerly Analytical Spectral Devices, at IS, CY and TR) and a PSR + spectrometer (Spectral Evolution Inc., Lawrence, Massachusetts, USA, at GR, AL, FY, BG, EG, and RS), respectively. In order to ensure that the machine learning models using the VIS-NIR-SWIR spectral datasets, that were measured in different laboratories using different spectrometers, produced valuable results, we applied a standardization protocol as introduced by Kopačková and Ben-Dor (2016), which uses an internal soil standard. In this context, we performed the measurements inside a dark box under artificial illumination to be independent of natural conditions. We recorded and averaged five spectra measurements for each soil sample in order to give its reflectance. A Spectralon® panel with 99% reflectance was used to calibrate the spectrometer prior to the measurements. Further details on the methodology are described by Ben Dor et al. (2015).

Subsequently, the measured spectra were down scaled to a resolution of 5 nm, due to a strong collinearity. Then, widely employed scatter corrective and spectral derivation pre-treatment techniques were applied for the dataset pre-processing stage, as summarized in Table 2. In brief, the reflectance spectra (Ref) were converted into pseudo-absorbance spectra [$\log_{10}(1/R)$] (Abs), and transformed into a continuum removal method domain (Abs-CR; Gomez et al., 2008). In addition, the Savitzky and Golay (1964) filter smoothing combined with standard normal variate transformation (Barnes et al., 1989) was used as pre-treatment before the spectroscopic modelling to remove unwanted background noise from the spectra (SG), and additionally the 1st derivative was applied (SG-1D). Supplementary, pre-processing techniques such as the de-trending (DT; Barnes et al., 1989), first (Abs-1D) and second derivative (Abs-2D) of the absorbance spectra and the multiplicative scatter correction (MSC, Geladi et al., 1985) were calculated for the rest of the analysis. An overview of those techniques is presented by Rinnan et al. (2009).

2.3. Dataset selection and partition

Considering the large differences in the spectral response of both mineral and organic soils in the entire dataset, the soil samples were divided in response to the values of the two above categories prior to the analysis, according to the Food and Agriculture Organization of the United Nations definition (organic soils are considered soils with > 20% of organic carbon content, Working Group WRB, 2015). Consequently, eight organic soil samples were identified and then removed from the spectroscopic analysis in order to focus on the mineral category.

In addition, we identified 31 outliers from the spectral attributes (Fig. 2). These were then removed from the SSL before the spectroscopic modelling. For the identification of the outliers, we followed the

Table 2
Spectral pre-treatments' abbreviations and description.

Abbreviation	Description
Ref	Reflectance values
Abs	Pseudo-absorbance values
Abs-CR	Continuum removed spectra of the absorbance values
SG	Applying the Standard Normal Variate transformation and a Savitzky-Golay filter of differentiation order 0, polynomial order 3 and window size 101
SG-1D	First derivative of the absorbance applying the Standard Normal Variate transformation and a Savitzky-Golay filter of differentiation order 1, polynomial order 3 and window size 101
DT	Applying the de-trending technique to the absorbance values
Abs-1D	First derivative of the absorbance
Abs-2D	Second derivative of the absorbance
MSC	Applying the Multiple Scatter Correction to the absorbance values

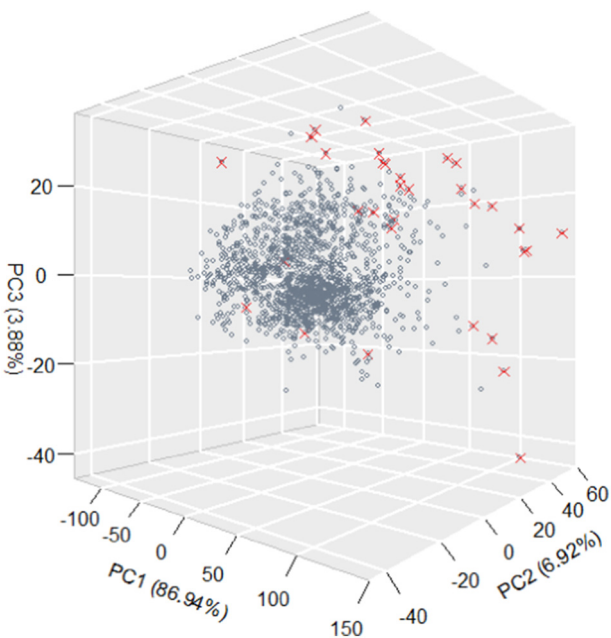


Fig. 2. Position of the spectral outliers (red crosses) in the 3D space of the 3 first PCs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

approach described by Filzmoser et al. (2005). First, the soil spectra were analysed by Principal Component Analysis (PCA). The six first Principal Components (PCs) were retained since they described over 99.5% of the information of the GEO-CRADLE SSL. Then, we used the Mahalanobis distance to calculate the distance of each soil sample on the projected space, whose squared values follow approximately a chi-square distribution with six degrees of freedom (which is equal to the number of PCs used). The selection was made using a critical threshold of 99.5% after fitting a cumulative distribution function of χ^2_6 .

The remaining 1721 soil samples were independently partitioned for each soil property. Each national SSL was divided into samples of calibration (80%) and validation (20%) set uniformly by the Kennard-Stone algorithm (Kennard and Stone, 1969), so that the combined calibration set was distributed evenly based on spectral representativeness, containing samples from each country and covering the extent of the soil mineralogy. Then the respective calibration and validation sets of the national datasets were merged to form a unified calibration and validation set.

2.4. Spectroscopic modelling

2.4.1. The modified local Gaussian regression approach

Whereas global approaches construct a single model using all available patterns from the calibration set, local ones establish several models each using a subset of the calibration patterns. These subsets are created usually by considering spectral, spatial, or pedological similarities. In this context, Ramirez-Lopez et al. (2013) proposed the SBL algorithm, which is a special kind of local model within the domain of memory-based learning. SBL constructs a single model for each testing pattern, by involving only its k closest neighbours from the calibration set according to a spectral similarity. This similarity is the Mahalanobis distance, calculated in the Principal Component space, using an optimal number of components. The model then performs regression using a Gaussian process which, in addition to the neighbours' spectra, also considers their pairwise distances as predictors. On the other hand, Nocita et al. (2014) developed their own memory-based learning approach, termed local-PLS, which uses the PLS algorithm to perform regression. Instead of relying solely in the spectral distance for the

selection of neighbours, local-PLS also involves the spatial and textural similarity.

The modified LGR algorithm proposed herein is a memory-based learning approach incorporating attributes of both SBL and local-PLS and extending them by proposing for the first time the use of combined spectral sources. LGR works by selecting the closest neighbours using both spatial and spectral similarity, with the latter calculated using the spectra derived from pre-processing technique P1. It then performs regression by using as predictors the spectra of the neighbours calculated using a different pre-processing technique P2, as well as their local pairwise distances. In that regard, it strives to maximize the complementary information contained between two different pre-processing techniques (referred to hereafter as combined spectral sources model), instead of relying solely on the best spectral pre-processing technique (single spectral source model).

More formally, LGR is a kind of memory-based learning method in which a Gaussian algorithm is used to construct local regression models for each unknown soil sample (u_i) based on suitable nearest neighbours selection from a set of reference samples.

Assume a set of n reference samples $(Xr, Yr) = \{x_{rj}, y_{rj}\}_{j=1}^n$, where Xr and Yr denote the spectra and corresponding soil attributes of the reference samples, respectively. For each validation sample u_i , its k closest neighbours from the reference set are selected, and the regression technique is performed using these samples to predict the soil properties of the targeted u_i . The method of LGR involves the following steps:

1. Fit a pair of pre-processing techniques ($P1, P2$) in spectra and derive spectral spaces XrA and XrB , respectively. Compute the distance vector ($d_{i,j}$) between u_i and each j reference sample; the distance is calculated as a combination of the spectral distance (calculated in spectral space XrA derived from the first pre-processing technique, $P1$) and of the geographical distance.
2. Generate the most similar local calibration sets to u_i as the k -nearest neighbours by minimizing each $d_{i,j}$ as generated in step 1;
3. A Gaussian process regression with a linear covariance function is performed for each u_i using the k -nearest neighbours defined in step 2. The LGR uses as predictors the spectra in space XrB defined by the second pre-processing technique ($P2$) and the pairwise distances calculated in spectral space XrA .

A summary of the steps aforementioned is illustrated in Fig. 3.

2.4.2. Distance calculation

The distance calculation between an unknown sample u_i and the j -th reference sample involves the combination of two similarities: on the one hand the spectral similarity, and on the other hand the geographical proximity.

As far as the spectral distance ($dS_{i,j}$) is concerned, this is calculated in the optimized principal components space as detailed in the SBL algorithm (Ramirez-Lopez et al., 2013), using the spectra formed in the first of the pre-processing techniques used.

The distance vector (d_i) is computed as a weighted mean of the spectral distance vector and the geographical coordinates as additional covariate distance vector (Eq. (1)).

$$d_{i,j} = w \cdot dS_{i,j} + (1 - w) \cdot dC_{i,j} \quad (1)$$

where $w \in [0, 1]$ and denotes the weight, while $dS_{i,j}$ and $dC_{i,j}$ are the spectral and geographical coordinates distance vectors of sample i , respectively. The distance vectors aforementioned are scaled prior to the computation of Eq. (1) by min-max normalization, and the geographical distance is calculated using a Euclidean weighted distance metric.

2.4.3. Parameter tuning and model development

Overall, our LGR approach involves the following parameters that should be optimized: the number of principal components for the

optimized principal components distance matrix computation, both pre-treatment techniques, the number of nearest neighbours for the local regression, and finally the weight (w) among the co-variables.

We evaluated three possible w with corresponding values of 0, 1/2 and 1, respectively. Then, we determined the optimal number of k by testing values from 40 to 250 with a step of 15. Moreover, we identified the optimum number of PCs as the ones producing a model having the smallest value within one standard error of the minimal RMSE of cross-validation, by applying a local leave-group-out cross validation for the internal validation procedure. The global RMSE was taken as the average of the RMSE of all local models. The optimal pair of spectral pre-treatment was experimentally defined among the techniques as detailed in Section 2.2.2. That is, each combination of the aforementioned nine spectral datasets was tested ($9 \times 9 = 81$ different combinations) and the one minimizing the RMSE was selected as optimal through a grid search using the internal cross-validation procedure detailed above.

Of the remaining steps required for the local calibrations we follow the recommendations described by Ramirez-Lopez et al. (2013). Therefore, and since we are dealing with high-dimensional spaces, a Gaussian process regression has been selected for the calibration model. Moreover, the inclusion in each local model of the local distance matrix among the neighbours (a $k \times k$ matrix extracted from the optimized principal components distance matrix, spatio-spectral combination) as a source of predictor variables was investigated experimentally to account for the complexities of the VIS-NIR-SWIR space.

2.4.4. Alternative global and local modelling approaches

In order to demonstrate the performance of the LGR model, state of the art machine learning algorithms for regression analysis were deployed with the same pre-processed spectra as the LGR model. These algorithms were:

The classical PLS algorithm (Geladi and Kowalski, 1986) which has been used for different purposes in soil spectroscopy. The PLS algorithm transforms the input factors' matrix by selecting latent variables (LVs) to maximize the covariance among the predictors and target parameters. In the current study, we restricted the number of LVs to 40.

The L-PLS algorithm (a modified version of the LGR algorithm), which uses optimized principal component as source of additional predictors and generates local PLS models to derive the predictions (it should be mentioned that it is a different approach compared to local-PLS; Nocita et al., 2014). In our implementation, we tested the L-PLS by selecting a leave nearest neighbour out cross validation for the internal validation and PLS for the local multivariate regression.

2.4.5. Evaluation of the models

The RMSE of the validation dataset (Eq. (2)), the ratio of performance to deviation (RPD, Eq. (3)), the coefficient of determination (R^2 , Eq. (4)) and the ratio of performance to interquartile range (RPIQ, Eq. (5)) were used to assess the performance of the local regression models. The RPIQ is commonly used as a better indication for non-normal variables (Bellon-Maurel et al., 2010).

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}} \quad (2)$$

$$RPD = \frac{SD_y}{RMSE} \quad (3)$$

$$R^2 = \left(1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \right) \quad (4)$$

$$RPIQ = \frac{IQ}{RMSE} \quad (5)$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value of sample i , and N the number of observations (Eq. (2)), SD_y is the standard

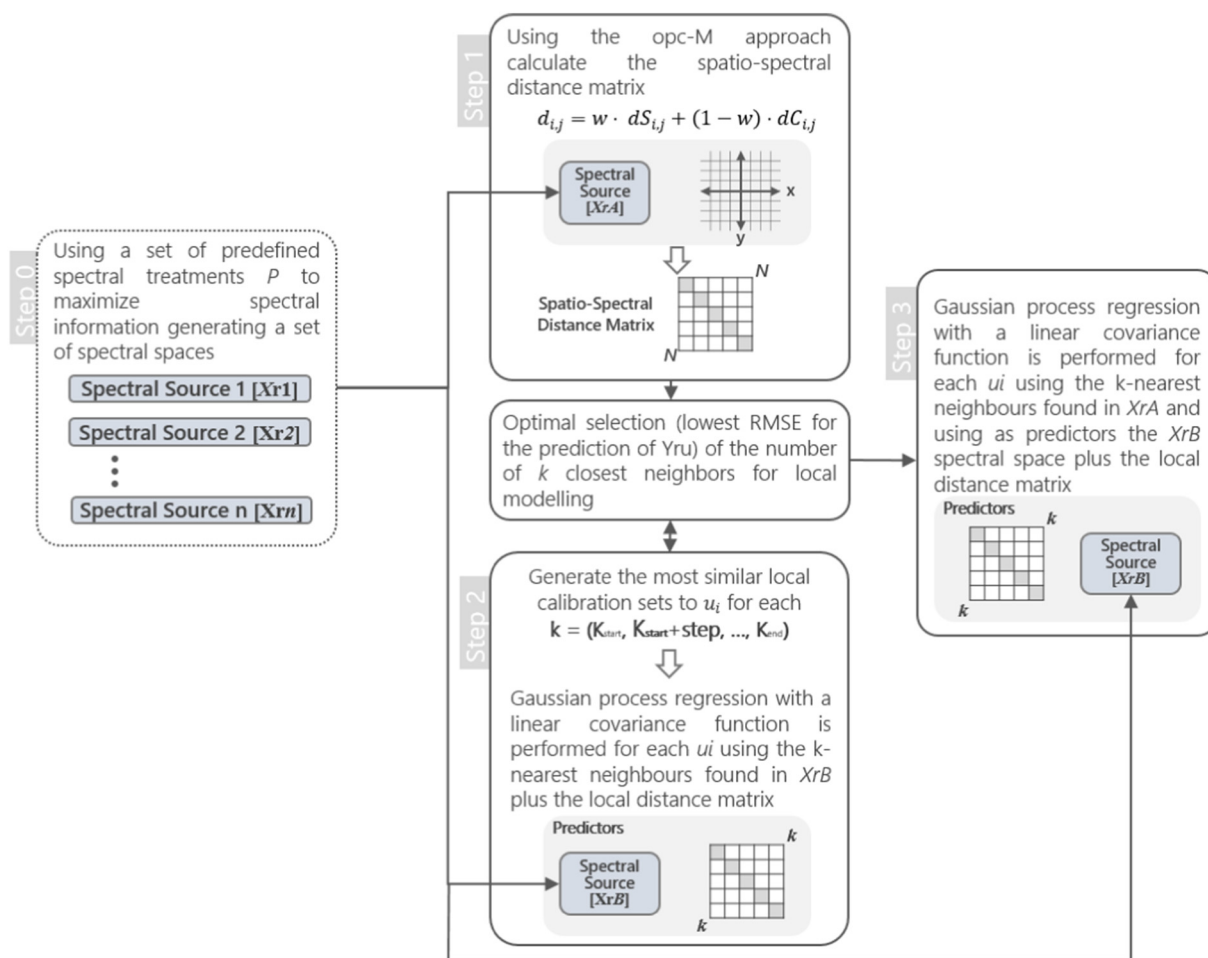


Fig. 3. Description of the steps carried out in the LGR approach; XrA and XrB are the spectral spaces as derived by the application of pre-processing techniques P1 and P2, respectively. opC-M refers to the optimized principal components distance.

deviation of the observed values (Eq. (3)), and IQ is the interquartile range ($IQ = Q3 - Q1$) of the observed values (Eq. (5)).

2.5. Implementation

The above mentioned spectroscopic and statistical analyses were performed utilizing the R software (R Development Core Team, 2013), with the resemble package (Ramírez-Lopez and Stevens, 2016). The caret package (Kuhn, 2008) was also used for the development of the competing methodologies to the LGR approach.

3. Results

3.1. Main characteristics of the GEO-CRADLE SSL

The GEO-CRADLE SSL is characterized by a wide diversity, comprising soils of 18 soil classes, as illustrated in Fig. 4. A 0.34% of the soil samples have no records of their soil class. We use in the following the classification system as proposed by the World Reference Base (WRB) for soil resources (Working Group WRB, 2015). The predominant soil groups are the Fluvisols, and the Cambisols that in general cover a large part of the Balkan and Mediterranean region (Zdruli et al., 2011). However, it is worth noting that these soil types are also the predominant soil groups in the Greek SSL containing the majority of the samples.

The classes aforementioned belong to soils with higher sand content and lower silt and clay as illustrated in the GEO-CRADLE SSL soil texture triangle (Fig. 5). Of all the GEO-CRADLE SSL soil samples, 99%

have clay measurements and 87% have measurements in silt and sand content, while Egypt and the Republic of Serbia have no records for silt and sand, hence their values were not illustrated in the texture triangle.

The soil samples originate from diverse territories and from top-soil down to deeper horizons, thus a wide range of variation was recorded across the observed values of the mineral soil samples. Table 3 presents the GEO-CRADLE SSL data and their descriptive statistics that were used in the modelling approach. Overall, GEO-CRADLE SSL exhibits low average OC contents (1.23%). Previous studies have confirmed these OC values for the countries along the Southern Eastern Balkan and North Africa countries (Jones et al., 2005; Henry et al., 2009). The average value for $CaCO_3$ and EC is 5.96% and 0.47 mS/cm, respectively while pH is considered neutral (mean = 7.17). With the exception of pH and silt, all other properties have positively skewed distributions. (Table 3). It should be noted that the statistical distributions of EC and $CaCO_3$ were strongly positively skewed (skew > 3; Table 3), therefore these were transformed ($\log_{10}(x + 1)$, where x is the variable to be transformed), to approximate normal distributions.

3.2. Spectral properties of the GEO-CRADLE SSL

In order to illustrate the soil spectral variability, we plotted the average Ref and Abs-CR spectrum, as well as their standard deviation per country (Fig. 6). In general, the Abs-CR spectra of soil samples indicate a large soil mineralogical variation in the GEO-CRADLE SSL, mainly due to the diverse soil-forming environments of the sample sites, human practices and their different soil layers. For instance, it can be clearly seen that soils from Egypt and Turkey resulted at lower

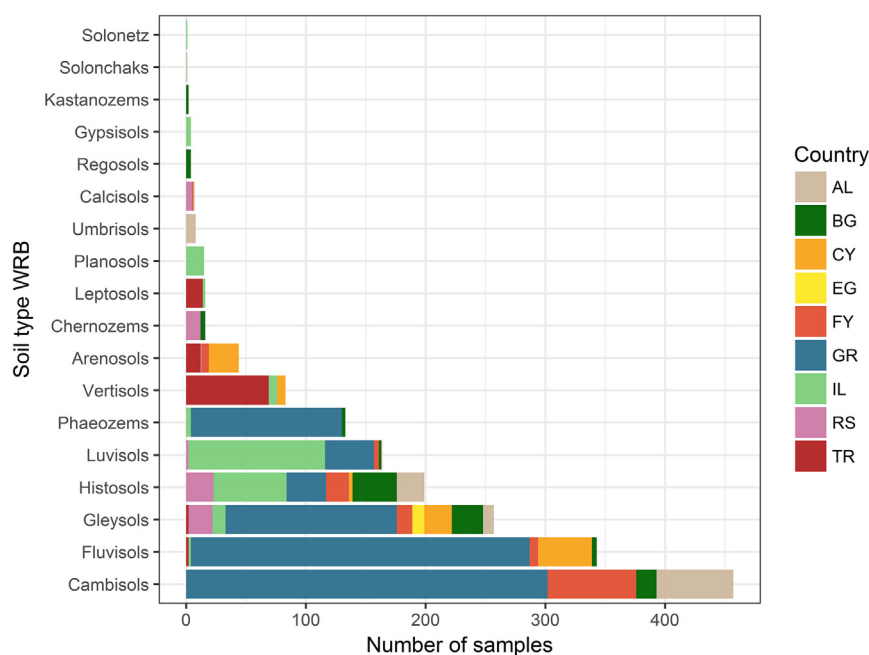


Fig. 4. Histogram of soil classes of the samples in the GEO-CRADLE SSL.

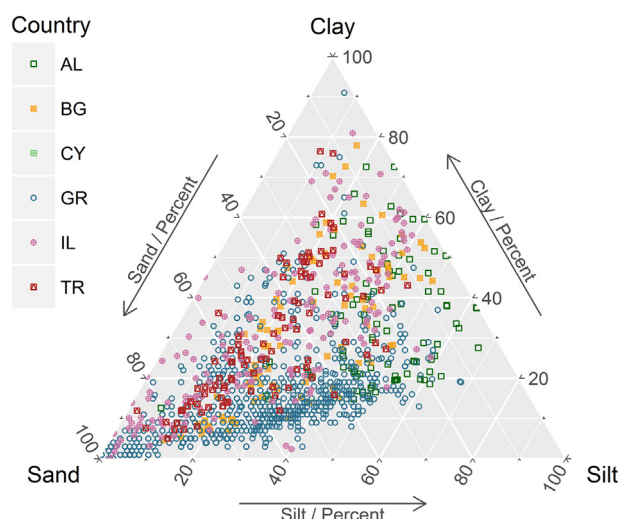


Fig. 5. Texture distribution of the samples in GEO-CRADLE SSL.

reflectance values in comparison to soils from Bulgaria and Greece since the samples were collected from the top layer in which the OC content displays higher values than in the deeper layers. The spectral variation observed across the different countries may justify the use of spatial proximity in the regression analysis. Significant absorption peaks can

also be seen among all samples around 1420 and 1915 nm caused by the overtones of O–H and H–O–H stretch vibrations of free water (Ben-Dor and Banin, 1995).

The three leading principal components of the GEO-CRADLE SSL spectra (after the outlier removal) described 98% of the information and their plots of scores are illustrated in Fig. 7. The remaining components represent only small proportions of the variance in the spectra and so they were not illustrated here. It can be clearly observed that the soil samples from the same country could be potentially broadly clustered but are mainly scattered in the spectral data space. For instance, the subset of local nearest neighbours from a Bulgarian soil sample involve samples that are not necessarily representative in soil similarities, such as samples from Egypt and Israel.

The reflectance spectra and the pre-processed spectra of GEO-CRADLE SSL data are illustrated in Fig. 8. Overall, the selected treatment methods construct new feature spaces by pronouncing different regions in the spectral range and eliminating different effects. For example, DT spectra emphasize the differences in the visible range; the Abs-1D, Abs-2D and SG-1D transformations gave more prominence to the SWIR region alleviating any differences due to its albedo. In addition, the Abs-CR and DT had similar results, since both alleviate the difference between the baselines of the spectra. However, the pre-processed spectral signatures by Abs-CR showed larger variations than the DT at the start of the spectrum in the visible range (350–750 nm).

Table 3

Statistical summary of the mineral soil data after outlier removal in the GEO-CRADLE SSL.

Soil attributes	N	Min	Q25	Median	Mean	Q75	Max	Sd	Skew
OC (%)	1582	0.00	0.78	1.12	1.23	1.51	6.38	0.80	2.10
pH	512	3.50	6.43	7.40	7.17	8.01	10.07	1.12	−0.69
CaCO ₃ (%)	1320	0.00	0.00	0.10	5.96	2.00	89.99	14.72	3.03
logCaCO ₃	1320	0.00	0.00	0.04	0.35	0.48	1.96	0.54	1.56
EC (mS/cm)	333	0.02	0.00	0.12	0.47	1.52	7.97	1.01	4.15
logEC	333	0.01	0.00	0.05	0.12	0.19	0.95	0.17	2.23
Clay (%)	1699	0.00	9.15	17.00	21.35	30.07	91.00	15.92	1.09
Sand (%)	1491	0.40	36.11	54.00	53.22	69.00	99.00	23.12	−0.04
Silt (%)	1491	0.00	16.00	26.00	26.80	37.40	68.00	14.30	0.16

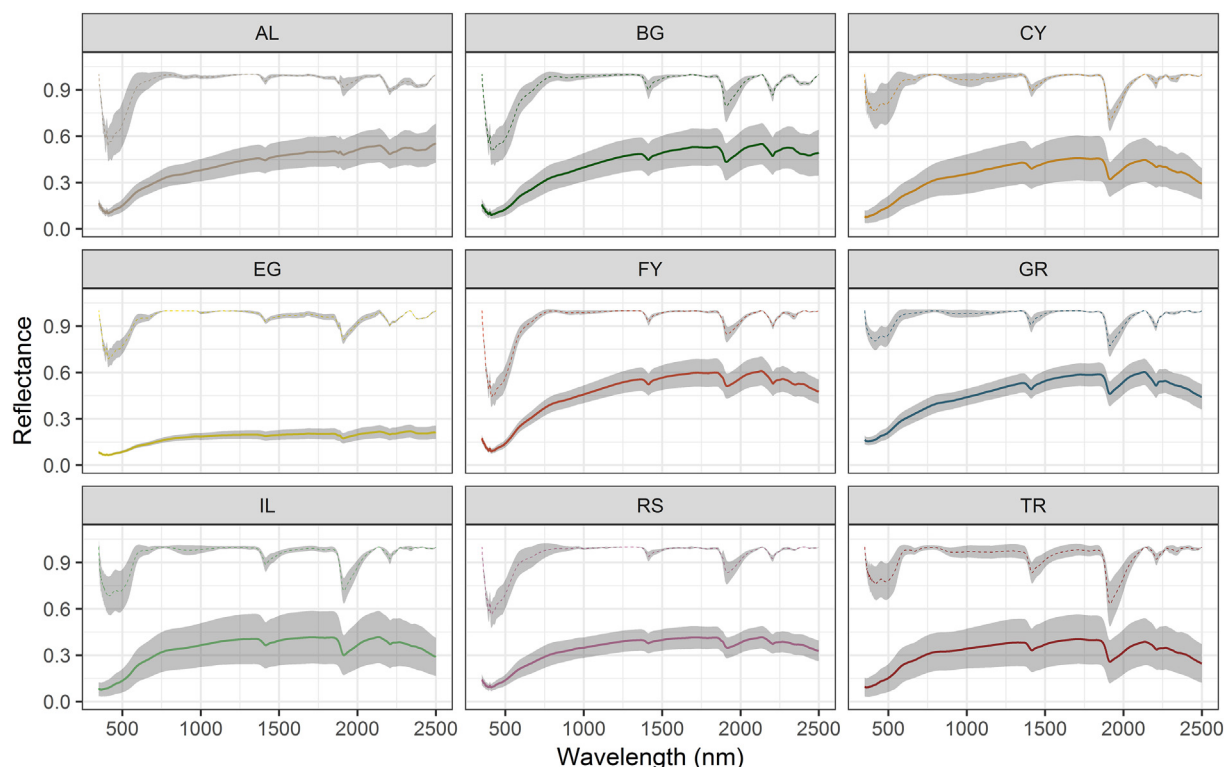


Fig. 6. Average reflectance (bottom) and continuum removed (top) spectra $\pm$ their standard deviation of the GEO-CRADLE SSL per each country.

3.3. Prediction of soil attributes

3.3.1. Effects of hyper parameter tuning

A summary of the main parameters for the best models is presented in Table 4. Overall, the number of k varied from 55 (for the model of logEC) to 220 (for the model of sand). It should be noted that k values strongly affected the predictive performance of all the soil properties. The PLS algorithm utilized Latent Variables (LVs) ranging from 13 to 39.

The advantage in LGR is that the incorporation of the geographical proximity in the distance metric computation enhanced the selection procedure of representative calibrations samples, since it overcame the problems caused due to the high diversity and imbalance of the soil samples. In order to better understand the influence of co-variables and their w on the soil properties prediction accuracy we illustrate the location of the 190 selected neighbours of a Cyprian soil sample (Fig. 9). The selection procedures were based on w using only the geographical coordinates (geo; Fig. 9a), only the spectral space (spec; Fig. 9b), and

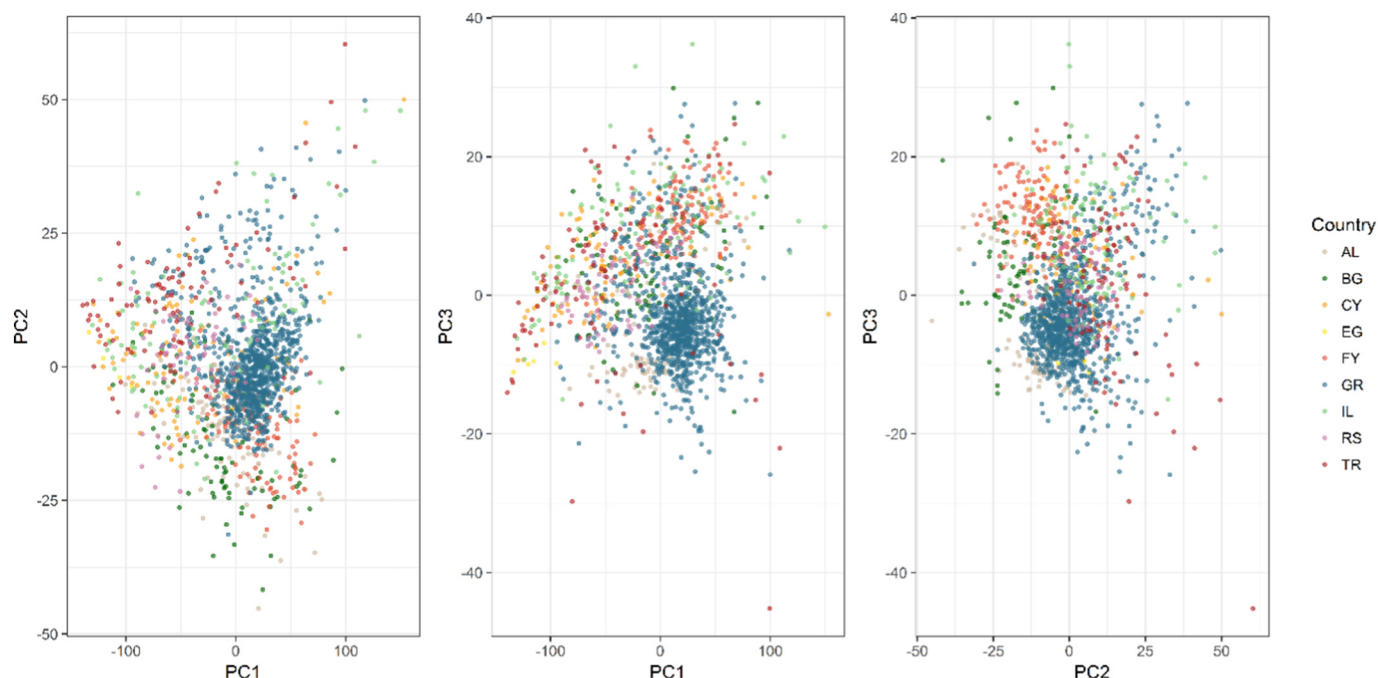


Fig. 7. Scatter diagrams of three leading PCs of the spectra data (after removal outlier), with points coloured according to their origin.

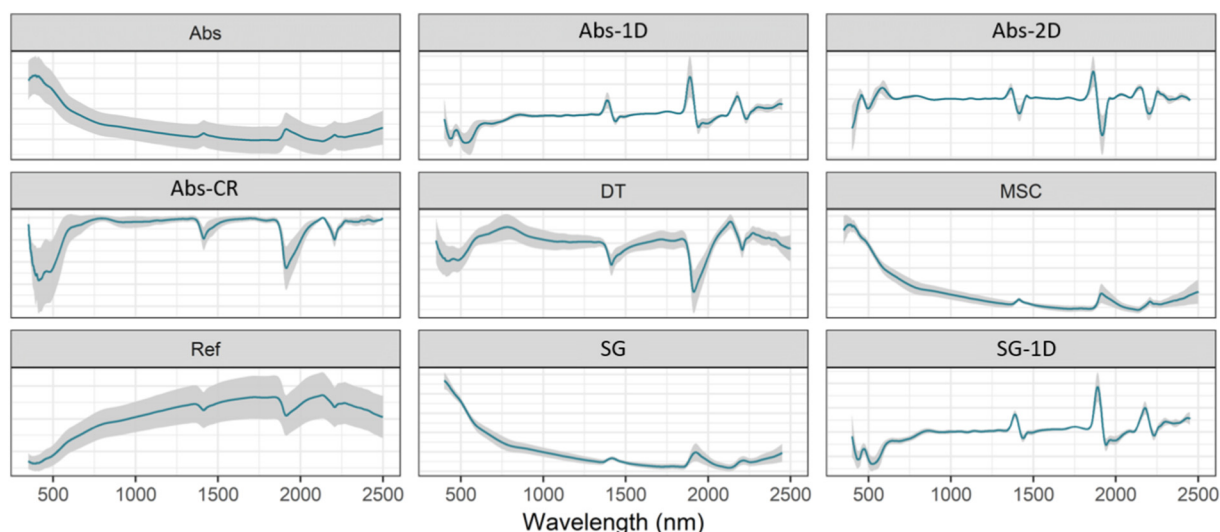


Fig. 8. The reflectance and the pre-processed spectra of GEO-CRADLE SSL data; The abbreviations for the spectral pre-treatments can be found in Table 2.

spectral and geographical space together (spec-geo; Fig. 9c). In this line the selection of the nearest neighbours based on combined spectral and geographical distances clearly had an influence on OC prediction and resulted in higher accuracy, compared to using solely spectral and geographical distances (spec-geo: RMSE = 0.30, RPD = 1.88, RPIQ = 1.89, $R^2 = 0.72$; spec: RMSE = 0.31, RPD = 1.51, RPIQ = 1.68, $R^2 = 0.58$; geo: RMSE = 0.33, RPD = 1.44, RPIQ = 1.59, $R^2 = 0.52$).

Moreover, in most cases shown in Table 4, the optimal pair of spectral pre-treatments of LGR is composed of different techniques – thus demonstrating that the use of combined spectral models, as an extension of single spectral models, can lead to results with enhanced prediction capabilities.

3.3.2. Comparison of predictive performances

In this section, we compare the assessment statistic values obtained by the proposed LGR approach with those from the PLS alone and the L-PLS. Based on these, LGR models show higher predictive performance than the other comparable algorithms (Table 5). Overall RPD (as derived by LGR) ranges from 1.68 to 3.66, R^2 from 0.64 to 0.93 and RPIQ from 0.69 to 4.72. It should be stated that low RPIQ values were generated by the EC and CaCO₃ due to their strong positive skewness and low interquartile range. In addition, it is notable that RPD (1.63 to 2.80), R^2 (0.62 to 0.87) and RPIQ (0.53 to 4.66) values for all L-PLS models are lower than those of LGR. Comparing the global PLS with local models it is obvious that PLS failed to model all variations in the training set resulting to lower predictive performance with its RPD values ranging from 1.37 to 2.68, R^2 from 0.48 to 0.85 and RPIQ from 0.36 to 2.48. In general, LGR generated accurate predictive models

since the RPD and RPIQ values are marginally close or quite substantially higher to 2.5, for the majority of the soil properties studied.

Also, we found moderate predictions for OC, EC and Silt ($1.5 < \text{RPD and RPIQ} < 2$). All LGR models also return errors that are substantially lower than that of the global PLS and L-PLS models. We illustrate the scatter plots of the best predictions in the independent test set across all soil properties, as produced by LGR which outperformed the other comparable algorithms (Fig. 10).

3.3.3. Effects of pre-processing methods on LGR

The optimal pair of pre-processing techniques (P) and the number of neighbours selected (k) of the best LGR models varied strongly among the different soil properties. Subsequently, performances of the pre-processing methods are compared for all experimental data sets, however only the models that use geographical coordinates and spectral space as auxiliary variables ($w = 1/2$; spec-geo) in the computation of distance are presented, since they concluded in higher prediction results.

The tests indicated that models utilizing different pre-processing methods in di computation and regression procedure (combined model) resulted in improved model performance. In contrast, the LGR models utilizing the same pre-processing in both steps (single model) had lower estimation accuracies. Actually, the single models follow the methodological framework of the SBL but in this case the geographical coordinates are also used in the computation of the dissimilarity metrics. Comparing the results of the combined and the single models demonstrates that different pre-processing methods and their combinations make LGR deliver accurate performance results (Table 6). Such results can be explained by the fact that the pre-processing methods (e.g., SG-

Table 4

Summary of the selected parameters of each algorithm; The abbreviations for the spectral pre-treatments can be found in Table 2. O-PCs refer to the optimized principal components.

Algorithm	Parameter	Soil properties									
		OC	pH	CaCO ₃	logCaCO ₃	EC	logEC	Clay	Sand	Silt	
PLS	LVs	39		22	27	27	34	13	35	24	34
	Treatment	SG-1D		DT	DT	DT	SG-1D	DT	Abs-1D	Abs-1D	SG-1D
L-PLS	k	145		115	175	85	70	70	175	190	170
	Treatment	DT + SG-1D		Abs-2D + SG	Abs-CR + SG	Abs-2D + SG	Ref + SG	Ref + Abs-1D	Abs-2D + SG	SG-1D + Ref	SG + Ref
	o-PCs	12		22	20	32	39	39	26	26	25
LGR	k	190		115	100	115	85	55	190	220	190
	Treatment	DT + SG-1D		Abs-2D + Abs-CR	SG-1D + Abs-2D	Abs-2D + Abs-CR	Ref + Abs-2D	Ref + Abs-CR	Abs-2D + Abs-CR	SG-Ref	DT-Ref
	o-PCs	12		22	19	32	39	39	26	24	27

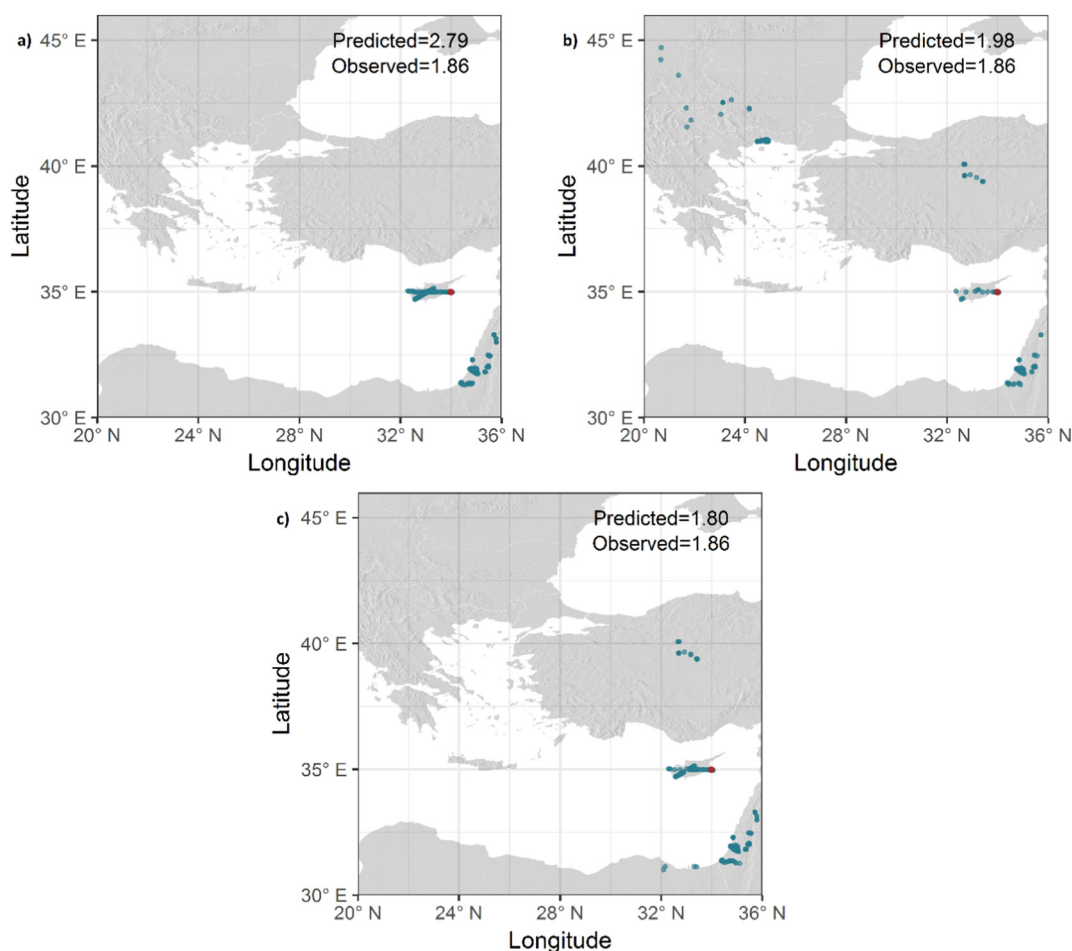


Fig. 9. Observed vs. predicted OC (%) values of a random prediction sample red (dot) and the locations of its selected predicted neighbours (blue dots) as generated using a) geographical b) spectral and c) spectral-geographical similarities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

1D, DT, Abs-CR) removed noise and enhanced signals and, hence, were beneficial for the selection of better neighbours in the spectral space. For instance, the prediction of clay was performed using Abs-2D in *di* computation and selection of the *k* nearest neighbours, while the Abs-CR treatment was utilized in the development of the prediction model. This combined model resulted to an RMSE of 6.32%, improving the single model's RMSE (6.63%). In overall, the predictions of the best single and of the combined LGR models for all the soil properties are depicted in Fig. 11, in which we can observe that the combined models improved the predictions by decreasing the RMSE by a relative 6.47%, ranging from 2.89% to 9.09%.

4. Discussion

Our results indicated that modelling of the GEO-CRADLE SSL with the modified LGR algorithm produced more accurate predictions for the soil attributes in comparison to the other local and global approaches (Table 5). One interesting feature which was introduced by the LGR is the use of different spectral treatments as a source of information in the computation of distance matrix and multivariate modelling carried out with the *k* nearest neighbours. This is a characteristic that it is not evaluated by any of the local algorithms applied in similar studies (e.g. Ramirez-Lopez et al., 2013; Nocita et al., 2014). The combination of different pre-treatment methods in all the cases improved the

Table 5
Assessment statistics of the global and local regression approaches.

Soil attribute	PLS				L-PLS				LGR			
	RMSE	RPD	RPIQ	R ²	RMSE	RPD	RPIQ	R ²	RMSE	RPD	RPIQ	R ²
OC (%)	0.40	1.37	1.25	0.48	0.35	1.77	1.69	0.68	0.30	1.88	1.89	0.72
pH	0.62	1.91	2.48	0.73	0.49	2.28	2.83	0.81	0.43	2.63	3.26	0.86
CaCO ₃ (%)	5.78	2.06	0.36	0.76	4.51	2.80	0.53	0.87	3.45	3.66	0.69	0.93
logCaCO ₃ (%)	0.20	2.68	2.45	0.85	0.22	2.53	4.66	0.84	0.22	2.56	4.72	0.88
EC (mS/cm)	0.56	1.56	1.06	0.59	0.45	1.80	1.10	0.69	0.43	1.93	1.12	0.74
logEC (mS/cm)	0.11	1.49	1.60	0.55	0.10	2.32	2.21	0.81	0.07	2.44	2.50	0.83
Clay (%)	9.90	1.58	1.58	0.60	6.89	2.28	2.35	0.81	6.32	2.48	2.56	0.84
Sand (%)	14.55	1.55	2.13	0.58	11.89	1.90	2.64	0.73	10.88	2.11	2.94	0.78
Silt (%)	10.18	1.41	2.03	0.50	8.76	1.63	2.33	0.62	8.52	1.68	2.39	0.64

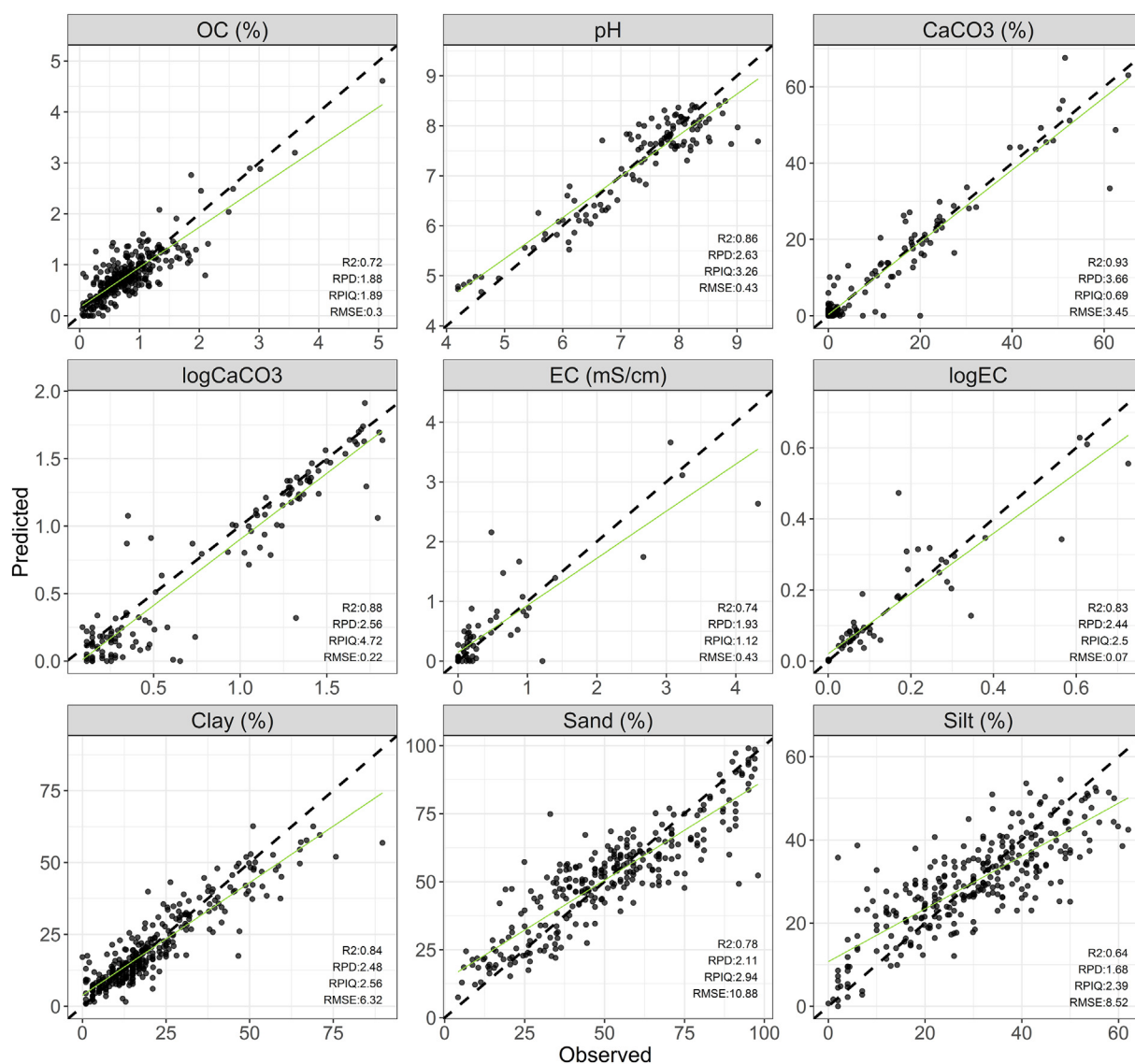


Fig. 10. Independent validations set of the best soil property predictions from LGR indicating the observed vs. the predicted values (green line represents the regression line while the dotted line represents the 1:1 line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

prediction accuracies of the modelling algorithms evaluated (Table 6; Fig. 11). In this line, it was clearly demonstrated that the various spectral pre-processing techniques result in complementary information which should be combined instead of relying solely on the best spectral pre-processing technique. In addition, we should highlight that the spectral standardization applied to reduce systematic and non-systematic spectral effects contributed to the overall improvement of the

predictive performance of the spectroscopic models. Advanced data handling and analytics techniques should be leveraged towards exploring multiple pathways for modelling spectral data and deriving more informative features. In that respect, Carmon and Ben-Dor (2017) relied on these techniques to develop automated data handling solutions for modelling of soil properties that allow the processing of the massive volumes of information arising out of pre-processed data,

Table 6

Comparison of combined and single predictive models. The abbreviations can be found in Table 2.

Soil attribute	OC	pH	CaCO ₃	logCaCO ₃	EC	logEC	Clay	Sand	Silt
Combined	DT + SG-1D	Abs-2D + CR	SG-1D + Abs-2D	Abs-2D + Abs	Ref + Abs-2D	Ref-CR	Abs-2D + Abs-CR	SG-Ref	DT-Ref
RMSE	0.30	0.43	3.45	0.22	0.43	0.07	6.32	10.88	8.52
RPD	1.88	2.63	3.66	2.56	1.93	2.44	2.48	2.11	1.68
RPIQ	1.89	3.26	0.69	4.72	1.12	2.50	2.56	2.94	2.39
R ²	0.72	0.86	0.93	0.88	0.74	0.83	0.84	0.78	0.64
Single	SG-1D	Abs-2D	SG	Abs-2D	Abs-CR	Ref	SG	SG	SG-1D
RMSE	0.32	0.46	3.55	0.24	0.45	0.08	6.63	11.31	8.93
RPD	1.63	2.47	3.55	2.35	1.81	2.13	2.37	2.01	1.60
RPIQ	1.62	3.07	0.67	4.33	1.05	2.18	2.44	2.80	2.28
R ²	0.64	0.83	0.91	0.82	0.69	0.80	0.82	0.75	0.61

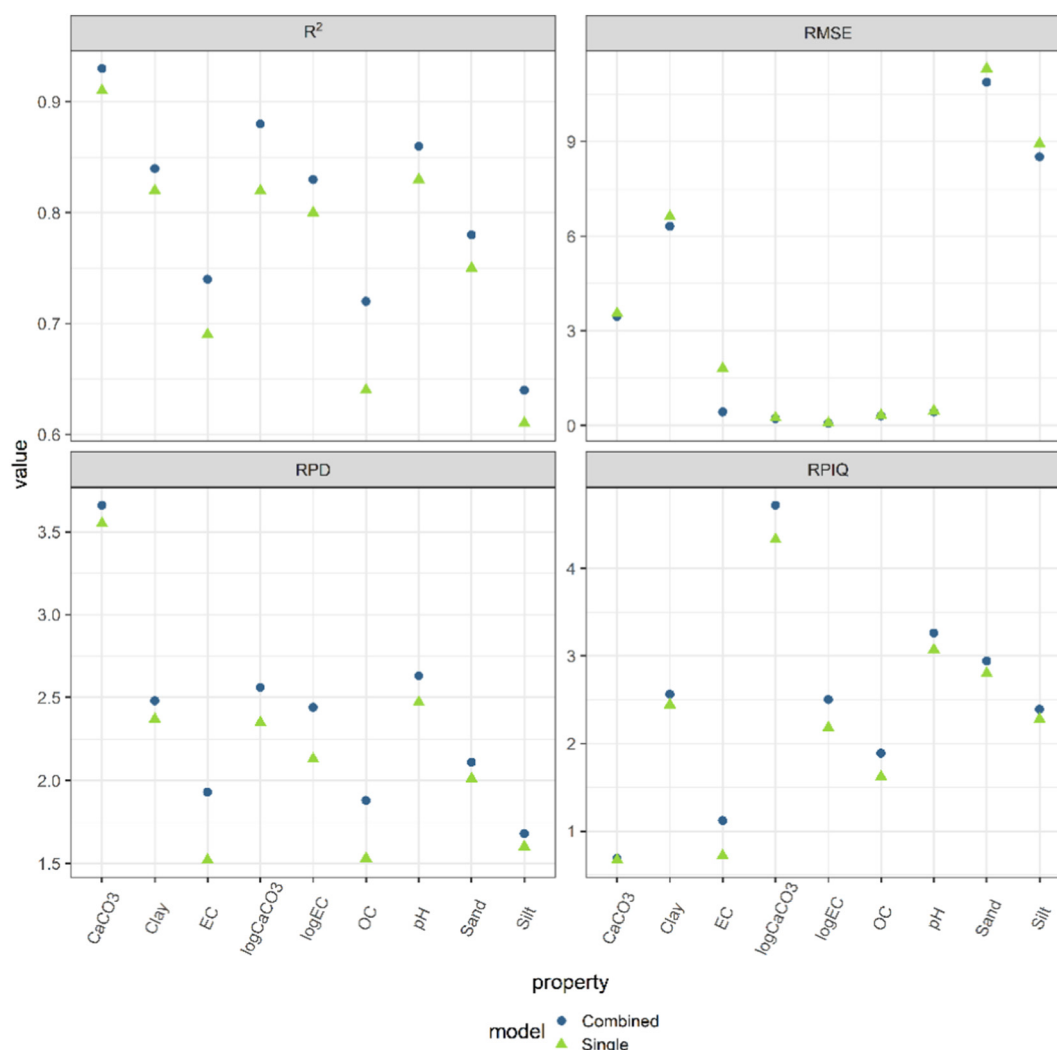


Fig. 11. Comparison of predictive LGR modes using a combined pair of spectral sources, and using the same single spectral source for both neighbour estimation and prediction.

towards the extraction of information related to spectral assignment explanation.

Taking into account that the information about the position of the soil sample contains some information about its compositional similarity, this could explain why the use of geographical coordinates as auxiliary variables in the computation of distances enhances the predictive power of models (Fig. 9). In this context, Shi et al. (2015) also showed that the incorporation of the geographical position into local calibrations models significantly improved predictions and it might be useful in modelling of large SSLs.

Other works have indicated that soil attributes from existing properties maps or data archives could be used as auxiliary variables in order to enhance prediction performance. For instance, Nocita et al. (2014) calibrated models of OC utilizing VIS-NIR data from LUCAS SSL combined with geographical coordinates and also sand content. However, in contrast to their approach, for the LGR, we did not evaluate other soil attributes as auxiliary predictors, hence we could not perform direct comparisons. From a scientific point of view, we think that the incorporation of soil attributes, such as pH as proposed by Brown et al. (2006) or clay content, constitutes a very promising approach which requires more research. Moreover, taking advantage of new technologies and latest developments in sensor science those measurements can be performed using cost efficient in situ sensors enabling a breakthrough in the efficient exploitation of SSLs in real field conditions.

Subsequently, there are many arguments for the selection of local regression approaches for improving the accuracy of soil properties predictions utilizing existing large SSLs as demonstrated in recent published studies (Lobsey et al., 2017). However, modelling an SSL with a local regression approach, such as our LGR, does not allow feature selection capabilities, due to the overall complexity derived by the local partitions. In this sense, global models (Tsakiridis et al., 2018) or more sophisticated fuzzy rule-based learning approaches (Tsakiridis et al., 2017b) are considered as a more favourable technique due to their interpretability. Finally, it is worth to mention that LGR, as any other local regression approach, selects a significant number of k-nearest neighbours needed for calibrating local regression models, corresponding to approximately 15% (ranging from 8% to 26%) of the reference set (Table 4). This is particularly relevant when modelling large SSLs (Nocita et al., 2014) and may raise issues in terms of computational time.

The development of GEO-CRADLE SSL facilitates further development of a global soil property and spectral database and supports regional contribution to global soil mapping activities and other regional GEOSS hubs, in an area with well recognized gaps. Last but not least, the importance of GEO-CRADLE SSL as an accurate source of data cannot be over-stated, in order to improve measurements, monitoring and reporting activities at various scales within the context of the Sustainable Development Goals (Anderson et al., 2017). Its role could

be highlighted as a proxy for estimating OC and produce large scale maps within the framework of indicator 15.3.1 and also might be useful to establish relevant policies. Considering the forthcoming hyperspectral satellite missions (Guanter et al., 2015), as well as the evolving technology of drones. Significant efforts are required to design effective strategies by integrating data from remote sensing data with ground-based measurements for soil properties estimation (Castaldi et al., 2018).

5. Conclusions

The results presented so far indicate that the proposed LGR model can be used to accurately estimate pH, CaCO₃, and soil texture components (clay, sand and silt) and fairly accurately estimate OC, and EC, which are crucial not only for agricultural functions but also for climate change mitigation/adaptation. We proved that spectral pre-treatments may be effectively combined enabling multivariate calibration approaches to yield enhanced accuracy results and accurately generate more relevant local calibration datasets. Furthermore, the addition of covariates, such as the geographical coordinates, in the computation of soil VIS–NIR–SWIR distance matrices in the local regression framework resulted in further improvements to the prediction performance of spectral models derived from a complex regional SSL.

Overall, the regional VIS–NIR–SWIR soil spectral library that we generated is a significant contribution to the global SSL initiative and can facilitate soil mapping activities in the Balkan, Middle East and North Africa region. The standardization process followed by the collaborating actors enable the achievement of promising results despite the high diversity of GEO-CRADLE SSL by means of samples origin, and measurement methods.

Future research could study whether these findings can be generalized to other soil attributes (e.g. bulk density, heavy metal content and micronutrients), and how this approach could be combined with spectra data recorded by proximal soil sensors. The SSLs that were generated in this project can be an excellent base to merge more regional SSLs that were measured under the same standardization process (e.g. the present Brazilian SSLs reported by Romero et al., 2018). Furthermore, the GEO-CRADLE SSL is being highly complementary with the recent research where hyperspectral remote sensing data are prioritised to assess the limitations whereby standalone multispectral satellite imagery is not sufficient to reach the desired representativeness and reliability for soil surface mapping. Soil spectroscopy research might therefore be focused on bridging the gap of the in-situ and space borne earth observation systems, putting specific emphasis on the upcoming launch of the next generation of imaging spectroscopy sensors.

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